

The combination of four main components in Xuebijing injection improved the preventive effects of Cyclosporin A in acute graft-versus-host disease mice by protecting intestinal microenvironment

Ting Shang^{a,b,1}, Yue Guo^{a,b,1}, Xiu-Rong Li^d, Zhengcan Zhou^{a,b}, Yubo Qi^c, Khalid Salahdiin^{a,b}, Ru Shen^{a,b}, Shuang He^{a,b}, Mei Wang^d, Zhe-Xin Shi^c, Xin Zhao^a, Jian Yang^{a,b}, Guanwei Fan^c, Yuefei Wang^a, Xiumei Gao^a, Yan Zhu^{a,b}, Yuxin Feng^{a,b,*}

^a State Key Laboratory of Component-based Chinese Medicine, Tianjin University of Traditional Chinese Medicine, No. 10 Poyanghu Road, West Area, Tuanbo New Town, Jinghai District, Tianjin 301617, China

^b Research and Development Center of TCM, Tianjin International Joint Academy of Biotechnology & Medicine, 220 Dongting Road, TEDA, Tianjin 300457, China

^c First Teaching Hospital of Tianjin University of Traditional Chinese Medicine, Tianjin, China

^d State Key Laboratory of Experimental Hematology, Institute of Hematology & Blood Diseases Hospital, Chinese Academy of Medical Sciences & Peking Union Medical College, Tianjin, China

ARTICLE INFO

Keywords:

Acute graft-versus-host disease
C0127
Chinese medicine
Intestinal microbiota
G-CSF

ABSTRACT

Acute graft-versus-host disease (aGVHD) is a major life-threatening complication after Allogeneic Hematopoietic Stem Cell Transplant (allo-HSCT). Although a series of immunosuppressant agents are routinely used as the first-line prevention, the morbidity and mortality rate remains high in allo-HSCT recipients. Our previous work indicated that combining Xuebijing (XBJ) with Cyclosporin A (CSA) is superior to CSA alone in preventing aGVHD. However, it was not clear which compounds in XBJ may prevent aGVHD. Whether the effective compounds in XBJ can be safely combined with CSA to prevent GVHD remain to be evaluated. Here, we accessed whether the combination of four main components in XBJ (C0127) had the same efficacy as XBJ in preventing aGVHD. In addition, the effectiveness of a novel combination therapy (C0127 + CSA) on aGVHD prophylaxis was evaluated using 16 s rRNA sequencing and RNA sequencing approaches *in vitro* and *in vivo*. In aGVHD mice, C0127 enhanced the preventive effects of CSA including decreasing mortality, maintaining weight, reducing GVHD score and reducing the expression of IL-6 and TNF- α in serum. Fatal GVHD is a frequent consequence of intestinal tract damage. We found combining C0127 with CSA alleviated the gut damage and maintained the normal physiological function of intestine by H&E staining, intestinal permeability and short chain fatty acid (SCFA) assays. Next, 16 S sequencing analysis of feces showed the combination treatment maintained the intestinal microbial diversity, normalized the intestinal microorganism and prevented flora disorder by reducing the relative abundances of *Escherichia coli* and *Enterococcus*. Further, RNA-seq analysis of colonic epithelium revealed C0127 combined with CSA chiefly regulated chemokines and cytokines in IL-17 signaling pathway. The combination treatment reduced the expression of G-CSF and its effector STAT3 (an axis that aggravated gut inflammation and flora disorder) in gut epithelium on mRNA and protein level. These findings indicated that C0127 improved the prevention of CSA in aGVHD mice partially by protecting the gut from damage through normalizing G-CSF signaling, which regulates the intestinal microbiota and the integrity of the epithelial barrier.

1. Introduction

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is a

well-established treatment for malignant tumors and blood diseases, which are incurable by conventional treatment. Acute graft-versus host disease (aGVHD) is a life-threatening complication after allo-HSCT.

* Correspondence to: State Key Laboratory of Component-based Chinese Medicine, Tianjin University of Traditional Chinese Medicine, 10 Poyanghu Road, West Area, Tuanbo New Town, Jinghai District, Tianjin 301617, China.

E-mail address: fengyn@live.com (Y. Feng).

¹ Equal contribution to this work.

<https://doi.org/10.1016/j.bioph.2022.112675>

Received 29 October 2021; Received in revised form 24 January 2022; Accepted 25 January 2022

Available online 17 February 2022

0753-3322/© 2022 The Authors.

Published by Elsevier Masson SAS. This is an open access article under the CC BY-NC-ND license

(<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Despite the existence of prophylactic immunosuppressive medication, the incidence of GVHD remains high [23,49]. Driven by activated donor T cells which attack host tissues, aGVHD primarily affects three organs: the skin, gastrointestinal (GI) tract, and liver [25]. The GI tract plays a critical role in GVHD [29]. Damage to the GI tract results in the amplification of systemic inflammation and dysbiosis (including loss of microbiota diversity), which leads to fatal GVHD and other unfavorable outcomes [3,19]. Recent preclinical and clinical studies suggest dysbiosis drives worse outcomes in allo-HSCT recipients. These results triggered a series of clinical researches to target the microbiota in allo-HSCT recipients [7,26,55].

The interleukin 17 (IL-17) family of cytokines has received much attention for its pro-inflammatory role in incorporating a complicated network of cytokines that is responsible for various inflammatory conditions and pathogenesis [36]. Accumulating evidences indicate that IL-17 has important roles in maintaining health during injury, physiological stress and infection [6,37]. As a key T helper cell population and therapeutic target in autoimmune disease, IL-17 also mediates late-onset skin GVHD and psoriasiform chronic GVHD [14,21]. Granulocyte colony stimulating factor (G-CSF), which is produced in response to interleukin-17A (IL-17A) synthesized by T cells that regulate neutrophils ($\gamma\delta$ and natural killer T cells). [13,30]. G-CSF widely acts in the bone marrow microenvironment to regulate haematopoietic cell proliferation and differentiation [10]. G-CSF-mediated inflammation can cause gut injury and flora disorder [1,63]. Besides, the secretion of G-CSF from intestinal epithelial cells influences intestinal homeostasis [22,44].

Xuebijing (XBJ) injection was prepared with extracts from five different Chinese herbs (Carthami flos, Paeoniae radix rubra, Chuanxiong rhizoma, Salviae miltiorrhizae radix, and Angelicae sinensis Radix). In 2004, it was approved by the China Food and Drug Administration (CFDA) to introduce into the clinical guideline of systemic inflammatory response syndrome (SIRS), multiple organ dysfunction syndrome (MODS) and sepsis [11]. It has been revealed that XBJ can reduce the release of inflammatory cytokines, enhance anti-oxidative stress ability, alleviate coagulation and improve microcirculation [20, 54]. Meanwhile, some researches revealed that four classes of compounds from five different herbs in XBJ may be important for its pharmacological effect including Honghua flavonoids, Chishao monoterpene glycosides, Chuanxiong/Danggui phthalides and Danshen catechols [38]. Intensive researches are going on to identify major active compounds in XBJ that can effectively treat sepsis [53,59,62].

Our previous studies revealed that combining XBJ with CSA yielded better outcomes in aGVHD preclinical model compared with CSA alone [40] (Zhou et al. [65]). In addition, we found the combination of four main compounds in XBJ (C0127) mimics the effects of XBJ. C0127 not only rescued mice from lethal *Candida* sepsis but also sustaining the survival of kidney epithelial cells [53]. Thus, we wondered whether the combination of four main compounds in XBJ (C0127) could also prevent aGVHD like XBJ when using together with CSA to identify the major active compounds of XBJ. In this work, it's prosperously that C0127 enhances the preventive effects of CSA in aGVHD mice. And we further indicated C0127 protected the gut tissues by reducing the G-CSF signaling to reverse the dysbiosis.

2. Methods

2.1. Chemicals and reagents

Xuebijing injection was standardized to contain 1.0–1.7 mg/mL of paeoniflorin and 0.2–0.5 mg/mL of hydroxysafflor yellow A as described [12,27]. C0127 [53,62], the combination of four main compounds in XBJ injection, was prepared with paeoniflorin, hydroxysafflor yellow A, ferulic acid and protocatechualdehyde in PBS according to reported concentrations in XBJ [39]. In detail, the concentrations of compounds in C0127 are paeoniflorin (1920 $\mu\text{g/mL}$), hydroxysafflor yellow A (580 $\mu\text{g/mL}$), ferulic acid (20 $\mu\text{g/mL}$), and protocatechualdehyde

(10 $\mu\text{g/mL}$). The four chemicals were ordered from Yuanye (Shanghai, China). All reagents used in the experiments were ordered from Sigma-Aldrich (St. Louis, MO, USA) unless specifically indicated. The Mouse Premixed Multi-Analyte Kit was purchased from Merck Millipore corporation (Billerica, Massachusetts, USA).

2.2. Experimental animals and ethical statement

This study was carried out in accordance with the recommendations of the Guide for the Care and Use of Laboratory Animals and the guidelines of Tianjin University of Traditional Chinese Medicine Animal Research Committee. The protocol was approved by the Tianjin University of Traditional Chinese Medicine Animal Research Committee. All transplantation experiments were performed with weight-(22–25 g) and sex-(male) matched 10 weeks BALB/c and C57BL/6 mice purchased from Vital River Company (Beijing, China). The mice in the study were housed in the pathogen-free animal facility and the overall health status was checked by trained professionals.

2.3. Acute GVHD model and bone marrow transplantation

A murine acute GVHD model was performed following the established protocol [2] (Zhou et al., [65]). Briefly, bone marrow (BM) cells were gently released from the femurs and tibias of donor C57BL/6 mice and suspended in PBS. Cell suspensions were then filtered through a 70 μm filter to obtain particulate free, single-cell suspensions. GVHD inocula were obtained from spleens of C57BL/6 mice. Recipient BALB/c mice were subjected to total body irradiation (4.25 Gy, twice, 4 h apart) the day before transplant (Day-1) via a Rad Source RS-2000 irradiator (San Diego, USA). Irradiated mice received donor BM (5×10^6 cells) and splenocytes (1×10^7 cells) by tail-vein injections on day 0. Mice were treated with 0.9% NaCl, or CSA (5 mg/kg or 2.5 mg/kg, intraperitoneally) alone, or C0127 (0.8 mL/kg subcutaneously) combined with CSA at the indicated time points. GVHD scoring was based on weight loss, posture, activity, fur texture, skin integrity, diarrhea and gut injury (severity score 0–2 for each variable, maximum index=12). Animals were euthanized if their weight loss > 35% or GVHD score > 7. The experiments were terminated on Day 18 after transplant.

2.4. Serum cytokines measurement

The blood was collected by removing eyeball of mice on Day 7, then centrifuged at 3500 rpm, 10 min to obtain the serum. The cytokines were detected using Mouse Premixed Multi-Analyte Kit following its described procedures.

2.5. Hematoxylin and eosin (H&E) staining

H&E staining was conducted as described [15]. Briefly, colon, liver, kidney tissues were collected on Day 7 after transplantation and were fixed with 4% formalin, dehydrated with alcohol, and embedded with paraffin. Then, the tissues were cut into 4 μm -thick sections, which were stained with H&E and visualized under a microscope (Vectra 3, PerkinElmer, USA).

2.6. The viability of CACO2 cells

The cell viability of CACO2 cell was measured using Cell Counting Kit-8. The cells were seeded in 96-well plate with 1×10^4 cells in each well. The adherent cells were incubated with C0127 diluted 20 and 100 times for 24 h before the damage caused by 10 μmol doxorubicin (DOX) for 12 h. Later, the viability of cells was measured according to the manufacturer's protocol on a FlexStation 3 microplate reader (Molecular Devices, Sunnyvale, CA, USA).

2.7. Real-time PCR

The total RNA was extracted using Trizol assay, including lysing cells, isolating/precipitating/washing RNA, appropriate ddH₂O was used to dissolve it, and then transcribed according to the standard methods and manufacturer's protocols of kit (Roche, Transcriptor First Strand cDNA Synthesis Kit). The qPCR Core Kit for SYBR Green (Germany) and the LightCycler 480 II (Roche) Real-Time PCR System were used as indicated by the manufacturer. The relative expression level of ZO-1 was calculated according to the $2^{-\Delta\Delta C_t}$ method. The relative mRNA level was determined using the comparative CT method and was normalized to the housekeeping gene glyceraldehyde-3-phosphate dehydrogenase (GAPDH). The primers were synthesized by Sangon Company (Shanghai, China). The primer sequences for real-time PCR were presented in Table 1.

2.8. Intestinal permeability assay

Intestinal permeability was assessed *in vivo* following oral administration of fluorescein-isothiocyanate (FITC)-dextran (7 kDa; Sigma). On Day 7, mice were orally gavaged with FITC-dextran (10 mg/20 g), four hours later, whole blood was obtained by cardiac puncture and centrifuged at 3500 rpm, 10 min. Serum was diluted with PBS in 1:1 and fluorescence intensity was measured using excitation at 490 nm and emission at 520 nm. The serum from mice that not received FITC-dextran was used as negative group.

2.9. Histopathological score of colons

The intestine pathological scores were based on the criteria in Table 2. The lymphocyte infiltration was observed at 200 × magnification [35].

2.10. Short-chain fatty acid analysis

Standard solutions: individual stock standard solutions of acetic acid, propionic acid, isobutyric acid, butyric acid, isovaleric acid, valeric acid, hexanoic acid, 2-ethyl butyric acid and 2-ethyl hexanoic acid were prepared by dissolving the analytes in 75% ethanol aqueous solution, the concentration of which were 23.58, 5.34, 0.46, 9.96, 0.70, 1.28, 1.04, 1.43, and 0.19 mg/mL. 2-ethyl butyric acid was employed as an internal standard for determination of acetic acid, propionic acid and butyric acid. 2-ethyl hexanoic acid was chosen as internal standard for quantitation of isobutyric acid, isovaleric acid, valeric acid and hexanoic acid.

Working standard solutions were prepared and serially diluted with 1% HCl and 75% ethanol aqueous solution containing 2-ethyl butyric acid and 2-ethyl hexanoic acid at a final concentration of 44.51 and

Table 2

The criteria for scoring colon injury of aGVHD mice, mainly include the following three aspects.

Score	Lymphocyte infiltration	Crypt aberrant	Colon wall aberrant
1	~ 10%	Basal 1/3 damage	The structure of wall is integrated
2	10–25%	Basal 2/3 damage	Colon wall thickening
3	25–50%	Crypt lost, surface epithelium present	Colon wall thickening, Invading into muscular layer
4	> 50%	Crypt and surface epithelium lost	Colon wall thickening, Invading into muscular layer with ulceration

2.26 µg/mL, respectively. The detail concentrations of 7 analytes were as following: acetic acid (33.16–1061 µg/mL), propionic acid (7.51–240.3 µg/mL), isobutyric acid (0.78–25.08 µg/mL), butyric acid (9.34–298.7 µg/mL), isovaleric acid (0.98–31.37 µg/mL), valeric acid (1.61–51.36 µg/mL), and hexanoic acid (0.98–31.26 µg/mL), respectively. Once prepared, the solutions were stored at –20 °C until analysis.

Sample solution preparation: As follows, the sample preparation procedure was optimized in terms of extraction solvent, extraction volume, the percentage concentration of HCl in sample solution, and ultrasonication time. Ultra-pure water (2.5 mL) was added to each 0.5 g fecal sample in a 15 mL tube, vortexed briefly, and then a total of 7.5 mL 1.33% HCl/ethanol solution containing internal standards were added to give the final concentration of 44.51 µg/mL (2-ethyl butyric acid) and 2.26 µg/mL (2-ethyl hexanoic acid). The samples were homogenized for 2 min, and centrifuged at 17,968 g for 10 min at ambient temperature. Supernatant was promptly transferred to a 2 mL sample vial.

GC analysis: Analysis was performed using a GC-2010 (Shimadzu, Kyoto, Japan) system equipped with an AOC-20i autosampler, and coupled to flame ionization detector. The chromatographic separation was performed on a DB-FFAP capillary column (30 m × 0.25 mm × 0.25 µm; Agilent Technologies, Palo Alto, CA, USA). A glass liner stoppered with a glass wool plug in the middle of the liner was used to prevent contamination of the GC column with nonvolatile fecal materials. Three microliters of the sample solution were injected in split mode at a ratio of 50:1. The injection temperature and detector temperature were 250 °C. The initial temperature of the column was 50 °C for 1 min, increased to 120 °C at a rate of 15 °C/min, to 170 °C at a rate of 5 °C/min, and then to 240 °C at a rate of 15 °C/min for 3 min. The total run time was 23.33 min. The carrier gas was N₂ (99.999%) at 1 mL/min. The make-up gas (N₂) was at 30 mL/min. The flow rate of H₂ and air were 47 mL/min and 400 mL/min, respectively.

2.11. 16S rRNA sequencing of fecal samples

The feces of the experimental mice were obtained in a sterile clean

Table 1
The primer sequences of the genes for real-time PCR.

Gene	Forward Primer (5'→3')	Reverse Primer (5'→3')
GAPDH	TGGTGAAGCAGGCATCTGAG	TGCTGTTGAAGTCGCAGGAG
IL13	CCTGGCTCTTGCTTGCCCT	GGTCTGTGTGATGTGCTCA
IL17RA	AGTGTTCCTCTACCCAGCAC	GAAAACCCGCCCGCTTAC
IL17RC	TTCTGCGGTATTGACTGTTTCG	GTCCCGGACTTCAAGACCC
G-CSF	ATGGCTCACTTTCTGCCACG	CTGACAGTGACCAGGGGAAC
CXCL2	CCAACCAACAGGCTACAGG	CGGTACACCTCAAGCTCTG
CXCL3	AGGCCCCAGGCTTCAGATAAT	AATGCAGGTCCTTCATCATGGT
CXCL5	TCCAGCTCGCCATTCATGC	TTGCGGCTATGACTGAGGAAG
S100A8	AAATCACCATGCCCCTCTACAAG	CCCACTTTTATCACCATCGCAA
S100A9	ATACTCTAGGAAGGAAGGACAC	TCCATGATGTCATTTATGAGGGC
LCN2	TGGCCCTGAGTGTTCATGTG	CTCTGTAGCTCATAGATGGTGC
CCL17	GACGACAGAAGGGTACGGC	GCATCTGAAGTGACCTCATGGTA
ZO-1(human)	AGGCGGATGGTGCTACAAGTGA	AGAGGACCGTGTAAATGGCAGACT
G-CSF(human)	TTAGAGCAAGTGAGGAAGATCC	CCATTCCCAGTTCTTCCATCT
STAT3(human)	TCCGCTAGAAACTGGATAACG	TGCAACTCCTCCAGTTTCTTAA

bench. 6 fresh stool samples of each group were collected in sterile tubes and frozen at -80°C . The genomic DNA of the sample was extracted by the CTAB/SDS method (Clark MS (1997) Plant molecular biology: A laboratory manual. Springer), then the purity and concentration of the DNA were detected by agarose gel electrophoresis. The genomic DNA was diluted to 1 ng/ μL with sterile water and used as a template for PCR (Novogene, Beijing, China).

16 S rRNA genes of distinct regions were amplified using specific primers with the barcode. All PCR reactions were carried out with Phusion® High-Fidelity PCR Master Mix (New England Biolabs, Ipswich, MA, USA). PCR products were mixed with the same volume of $1 \times$ loading buffer (contained SYB green) and separated by electrophoresis on 2% agarose gel. PCR products were mixed in equidensity ratios. Then, mixture PCR products were purified with GeneJET™ Gel Extraction Kit (Thermo Scientific, USA). Sequencing libraries were generated using Ion Plus Fragment Library Kit 48rxns (Thermo Scientific) following manufacturer's recommendations. The library quality was assessed on the Qubit®2.0 Fluorometer (Thermo Scientific). The library was sequenced on an Ion S5™ XL platform and 400 bp/600 bp single-end reads were generated.

Single-end reads were assigned to samples based on their unique barcode and were truncated by cutting off the barcode and primer sequence. The raw reads were performed under specific filtering conditions to obtain the high-quality clean reads according to the Cutadapt (V1.9.1, <http://cutadapt.readthedocs.io/en/stable/>) quality-controlled process [42].

Afterwards, the reads were compared with the reference database (Silva database, <https://www.arb-silva.de/>) [50] using UCHIME algorithm (UCHIME Algorithm, http://www.drive5.com/usearch/manu al/uchime_algo.html) [18] to detect chimera sequences, and then the chimera sequences were removed to obtain the clean reads [24].

Sequences analysis were performed with Uparse software (Uparse v7.0.1001, <http://drive5.com/uparse/>) [17]. Sequences with $\geq 97\%$ similarity were assigned to the same Operational taxonomic units (OTUs). Representative sequences for each OTU were screened for further species annotation in the Silva Database (Version 132) (<https://www.arbsilva.de/>) [50] based on Mothur algorithm to annotate taxonomic information. In order to study the phylogenetic relationship of different OTUs, and the difference of the dominant species in different samples (groups), multiple sequence alignment were conducted using the MUSCLE software (Version 3.8.31, <http://www.drive5.com/muscle/>) [16]. OTUs abundance information was normalized using a standard of sequence number corresponding to the sample with the least sequences. Subsequent analysis of alpha diversity and beta diversity were all performed basing on this output normalized data.

2.12. Ribonucleic acid sequencing (RNA-Seq) and data analysis

Total RNA was extracted from colon tissues and tested for purity and integrity. RNA purity was checked using the Nano-Photometer® spectrophotometer (IMPLEN, CA, United States). RNA integrity was assessed using the RNA Nano 6000 Assay Kit of the Bio analyzer 2100 system (Agilent Technologies, CA, United States). The library of transcriptome sequencing was then generated using NEBNext® Ultra™ Directional RNA Library Prep Kit for Illumina® (NEB, United States). After the library was constructed, Qubit2.0 Fluorometer was used for initial quantification, diluting the library to 1.5 ng/ μL , and the insert size of the library was detected with Agilent 2100 bioanalyzer. In addition, the library preparations were sequenced on an Illumina Novaseq platform. After obtaining clean data (clean reads) by deleting the reads containing adapter, reads containing ploy-N, and low-quality reads in the original data, these clean reads were compared to the Mus musculus genome utilizing the HISAT2 alignment software (The Johns Hopkins University, Baltimore, Maryland, United States). The quantification of gene expression level was then quantified using FPKM. Differential expression analysis of mice from different groups was performed utilizing the

DESeq2 R package (1.16.1). $\log_2\text{FoldChange} \geq 1$ and $P\text{-value} \leq 0.05$ were set as the threshold of significant difference expression.

2.13. KEGG enrichment analysis of differentially expressed genes

KEGG (Kyoto Encyclopedia of Genes and Genomes) is a comprehensive database integrating genomic, chemical and systematic functional information. We used cluster profiler R package to test the statistical enrichment of differential expression genes in KEGG pathways.

2.14. Western blotting (WB)

The colon tissue was treated with RIPA buffer (Solarbio, Beijing, China), and the supernatant was centrifuged at 13,000 rpm/min, 4°C for 10 min. Then the total protein concentration of the supernatant was measured with BCA Protein Assay Kit (PC0020, TransGen Biotech, Beijing, China). The SDS-PAGE protein loading buffer (P0015 L, Beyotime Biotechnology, Shanghai, China) was added to the supernatant, and the protein was deformed by heating in a metal bath at 100°C for 10 min. The proteins were separated by SDS-PAGE on 12% separation gels, transferred to hydrophobic polyvinylidene (PVDF) membranes, and blocked with TBST (Tris-buffered saline with Tween-20) containing 5% non-fat dry milk. The primary antibody (Rabbit G-CSF, 1:2000, bs-1023R, Bioss, Beijing, China) and GAPDH (1:4000, Cell Signaling Technology, Beverly, MA) was added dropwise to the PVDF membrane and transferred to 4°C refrigerator for 8 h. After washing with TBS-T, the secondary antibody (Goat anti-Rabbit IgG, 1:4000, ZB-2301, ZSbio, Beijing, China) was added to the PVDF membrane drop by drop and incubated at room temperature for 2 h, and then washed with TBST again. Finally, the EasySee Western Blot Kit (DW101-02, TransGen Biotech, Beijing, China) was used to react with the PVDF membrane for 1 min in darkness, and the bands were obtained by the gel imaging system. The protein expression of G-CSF was analyzed using Image J software (National Institutes of Health, Bethesda, MD, USA) and standardized to GAPDH.

2.15. Immunohistochemistry (IHC)

The paraffin sections of colon tissue were baked in an oven at 60°C for 2 h to melt the paraffin, then dewaxed in xylene and rehydrated. After inhibiting endogenous peroxidase activity with 3% H_2O_2 , the tissue sections were heated in 0.01 M sodium citrate buffer ($\text{pH} = 6.0$) and allowed to normal temperature. Non-specific binding sites were blocked with blocking buffer (10% bovine serum) for 1 h at 37°C . The sections were incubated with Rabbit anti-G-CSF polyclonal antibody (bs-1023R, Bioss, Beijing, China) diluted in blocking buffer (1:400) for 2 h at 37°C and washed twice with PBST for 5 min each, followed were sequentially stained using corresponding secondary antibody and third antibody for 40 min at 37°C . DAB kit (AR1022, Boster Bio, Wuhan, China) was used for color development, followed by counter staining with hematoxylin and differentiation with 1% hydrochloric acid alcohol. Finally, seal the sections with neutral gum after dehydration. The expression of G-CSF was detected by optical microscope (Vectra 3, PerkinElmer, United States) and quantified through the use of Image J software (National Institutes of Health, Bethesda, MD, USA).

2.16. Statistical analysis

The Log-rank test was used to determine the statistical significance of Kaplan-Meier survival curves. Other results were analyzed by unpaired t or ANOVA tests as appropriate, using InStat version 3.06 software for Windows (GraphPad, San Diego, CA, USA). The following terminology was used to show statistical significance: * $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$.

3. Results

3.1. C0127 enhanced the preventive effects of CSA in aGVHD mice

To test the regime outcomes of combining C0127 with CSA in aGVHD mice, we conducted the aGVHD model in BALB/C mice by tail vein injecting the splenocytes from C57BL/6 mice after total body irradiation (TBI), and then treating with corresponding medicines according to the schemes (Fig. 1A). We found C0127 combined with CSA further improved the survival of aGVHD mice compared with 5 mg/kg CSA alone (Fig. 1B). Simultaneously, the combination treatment could partly maintain the weight of aGVHD mice on Day 7, 8 and 9 after transplant (Fig. 1C and D). In addition, the GVHD scores were lower in combination regime group compared with aGVHD and 5 mg/kg CSA groups, which was evaluated based on uniform and agreed assessment criteria (Fig. 1E and F). In the early aGVHD phase, inflammatory triggers can drive both the innate and adaptive immune responses and inflammation plays a vital role in the initiation of aGVHD [5]. The cytokine array analysis revealed C0127 combined with CSA reduced the expression level of IL-6, TNF- α and IL-12 (Suppl. Fig. 1A–C) in serum collected on Day 7 after transplant, and compared with CSA, the combination treatment reduced the expression of IL-6 and TNF- α more significantly. These results suggested that C0127 enhanced the preventive effects of CSA in aGVHD mice.

3.2. Combining C0127 with CSA protected the intestinal tissues in aGVHD mice

The gastrointestinal tract is the most common target organ in pathological process of GVHD, we next explored whether the combination treatment could protect the gut from injury caused by TBI. Consistent with the above results, The H&E pathological section of mice in GVHD and CSA groups revealed disappeared crypts and broken structure of colon wall, however C0127 combined with CSA normalized the intestine tissues, especially in the crypts (Fig. 2A–D). And the histopathology scores of intestines were lower in mice treated with C0127 and CSA compared with the mice in aGVHD and CSA groups (Fig. 2E). Exposed to TBI, the intestinal epithelial barrier function would become incomplete,

we delivered the FITC-dextran into the systemic circulation by oral gavage on Day 7 after allo-HSCT to evaluate intestinal integrity by measuring the concentration of FITC-dextran in blood and found C0127 combined with CSA significantly reduced intestinal permeability (Fig. 2F). Metabolites produced by the gut microbiota play a crucial role in the control of the intestinal immune system, short chain fatty acids (SCFA) are among the most studied metabolites and are produced by the gut microbiota in the colon [34,51]. Therefore, we further measured the SCFA contents in feces using gas chromatography to evaluate the gut function and structure laterally, and the stool in C0127 combined with CSA group showed more quantity of butyrate, acetic acid and valeric acid compared with CSA alone (Fig. 2G–I), which further reflected the health of the intestinal system in mice treating with C0127 + CSA.

In addition, *in vitro*, we found C0127 improved the viability of intestinal epithelial cell (CACO2) and increased the number of viable cells when induced by Doxorubicin (DOX), a commonly used chemotherapy drug, cause damage to the body and organs (Fig. 2J, K). And Real-time polymerase chain reaction (RT-PCR) revealed that C0127 maintained the expression of tight junction protein 1 (ZO-1) in CACO2 cells induced by DOX (Fig. 2L). Therefore, we found the combined treatment protected the intestine from injury caused by chemoradiotherapy.

3.3. C0127 combined with CSA maintained the microbiota diversity in aGVHD mice

Based on the protective effects of C0127 + CSA in intestine, we hoped to test whether the combination treatment could normalize the microbiota via providing a healthy microenvironment for aGVHD mice. Firstly, we compared the diversity of the intestinal microbiome in mice treating C0127 + CSA with mice receiving CSA alone. The Alpha diversity indexes Shannon and Simpson in the microbiota of mice treating with C0127 + CSA are higher than CSA and GVHD groups (Fig. 3A and B), which reflected that C0127 combined with CSA could maintain the microbiota diversity in aGVHD mice. Principal Coordinate Analysis (PCoA) revealed that the distance of the samples between C0127 + CSA group and no-GVHD groups (Control and BT) was near, but the distance was far between C0127 + CSA group and CSA or GVHD group. The microbial structure or community of microbial in C0127 + CSA group

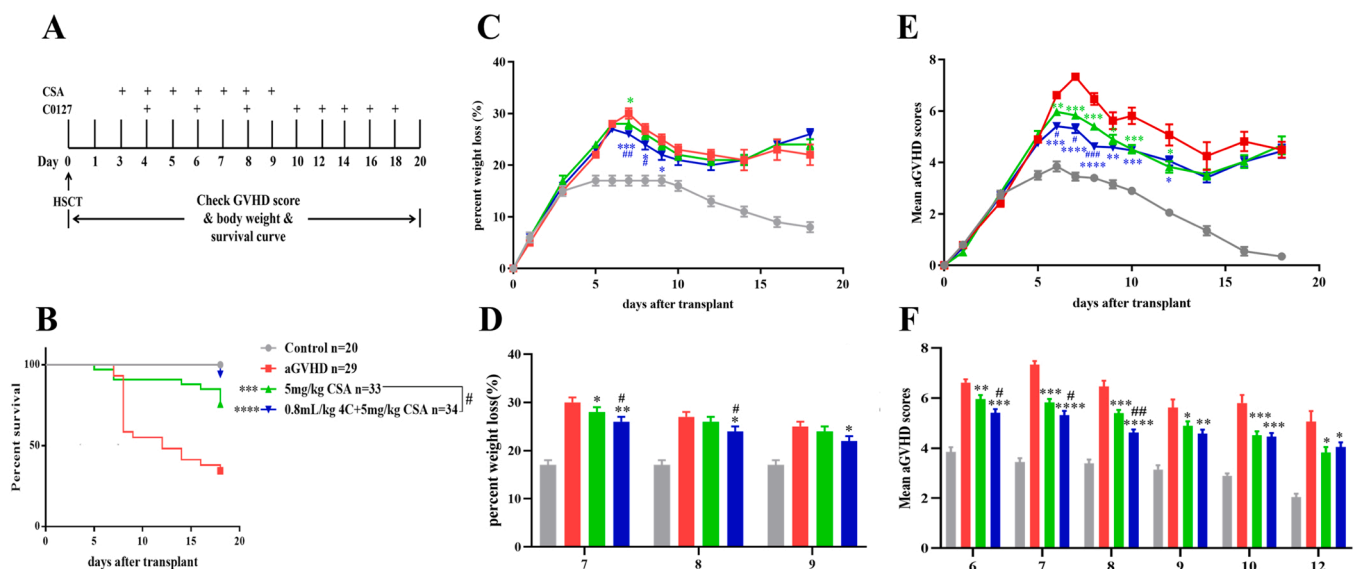


Fig. 1. C0127 enhanced the preventive effect of CSA on aGVHD mice. (A) aGVHD mice were treated with C0127 and CSA according to the experiment scheme. (B) The survival curve of aGVHD mice, which were injected with corresponding medicine. (C&D) Body weight loss after transplant in different groups of mice. (E&F) GVHD scores of mice in different groups, which were evaluated with strict scoring system. Animal numbers per group are depicted above panels. Survival was analyzed using the log-rank test. Other experiments were analyzed using ordinary one-way analysis of variance (ANOVA) for multiple comparisons. Data are means $\pm$ SEM or means $\pm$ SD. *** $P < 0.0001$, ** $P < 0.001$, * $P < 0.01$, # $P < 0.05$, compared with aGVHD; ## $P < 0.01$, ### $P < 0.001$, compared with 5 mg/kg CSA.

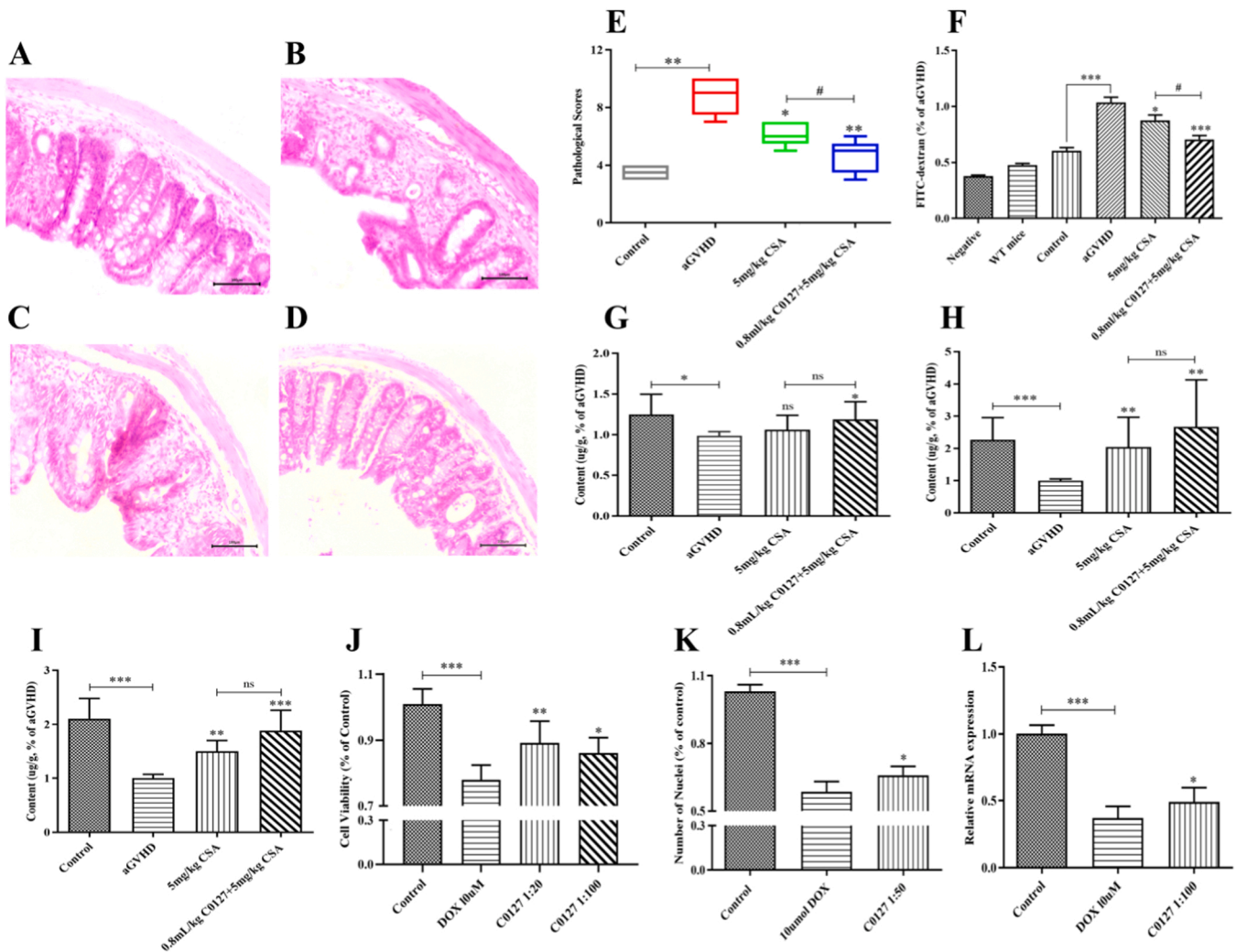


Fig. 2. C0127 prevented the intestine issues from damage caused by aGVHD and chemotherapy. (A–D) Representative images of hematoxylin and eosin-stained colons from aGVHD mice in different groups, (A) Control, (B) aGVHD, (C) 5 mg/kg CSA, (D) 0.8 mL/kg C0127 + 5 mg/kg CSA, scale bar = 100 μ m. (E) The pathological score of intestine ($n = 4$), the scoring criteria were shown in methods section. (F) FITC-dextran concentrations (% of aGVHD) in serum on Day 7 after transplant. (G–I) The contents of short chain fatty acids (SCFA) in feces, which was measured by gas chromatography, acetic acid (G), butyrate (H), valeric acid (I). Data are means $\pm$ SD and were analyzed using ordinary one-way analysis of variance (ANOVA) for multiple comparisons. *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$, compared with aGVHD; # $P < 0.05$, compared with 5 mg/kg CSA. (J–L) C0127 could protect intestinal epithelial cell (CACO2) from injury induced by doxorubicin (DOX). (J) The CCK8 assay was used to determine the viability of CACO2 cells. The cell were incubated with C0127 for 24 h, then injured with 10 μ M DOX for 12 h. (K) The nucleus number of CACO2, which were labeled by the Hoechst dye and high content analysis were used to detect the cell number. (L) The mRNA relative expression of ZO-1 (tight junction protein 1) in CACO2 cells, determined by real-time polymerase chain reaction (RT-PCR). In experiment of K&L, the management of cells were the same as J assay. Data are means $\pm$ SD. Data were analyzed using ordinary one-way analysis of variance (ANOVA) for multiple comparisons. *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$, compared with DOX 10 μ M group.

was similar to the Control and WT (BT) mice (Fig. 3C).

3.4. C0127 combined with CSA reversed abnormal microbiota in aGVHD mice at the phylum level

To further clarify the influence of the combination regimen of C0127 and CSA on microbiota of aGVHD mice, we analyzed the intestinal flora at phylum level. The heatmap of top 20 phylum showed the combination treatment partially reversed the dysbiosis in aGVHD mice (Fig. 4A and Suppl. Fig. 2A). Analyzing the top 10 phylum in abundance by MetaStat and UPGMA also revealed that the microbiota of mice undergoing combination treatment after allo-HSCT was similar to the no-GVHD mice (Fig. 4B and C). The box plots using MetaStat analysis presented the combination treatment maintained the abundance of Bacteroidetes to the level of no-GVHD mice (ATCON and BT) and reduced the Proteobacteria abundance (Fig. 4D and E). In agreement with our research,

some researches indicated that the Bacteroidetes are negatively correlated to the obesity and Bacteroides fragilis could increase gut microbiota diversity and significantly ameliorate acute and chronic GVHD development [28,56]. Meanwhile, Proteobacteria is frequently observed after allo-HSCT and outgrowth of the phylum Proteobacteria was also linked to increased incidence of GVHD, infections, and organ failure after allo-HSCT [57,60]. From above, we concluded that the combination treatment partly protected the intestinal flora from disorder in phylum by reducing the abundance of harmful bacteria and maintaining probiotics.

3.5. C0127 combined with CSA normalized microbiota of aGVHD mice at the species level

Next, we analyzed the 16 S RNA sequencing results on species level, aiming to identify the potential biomarkers of aGVHD and explored

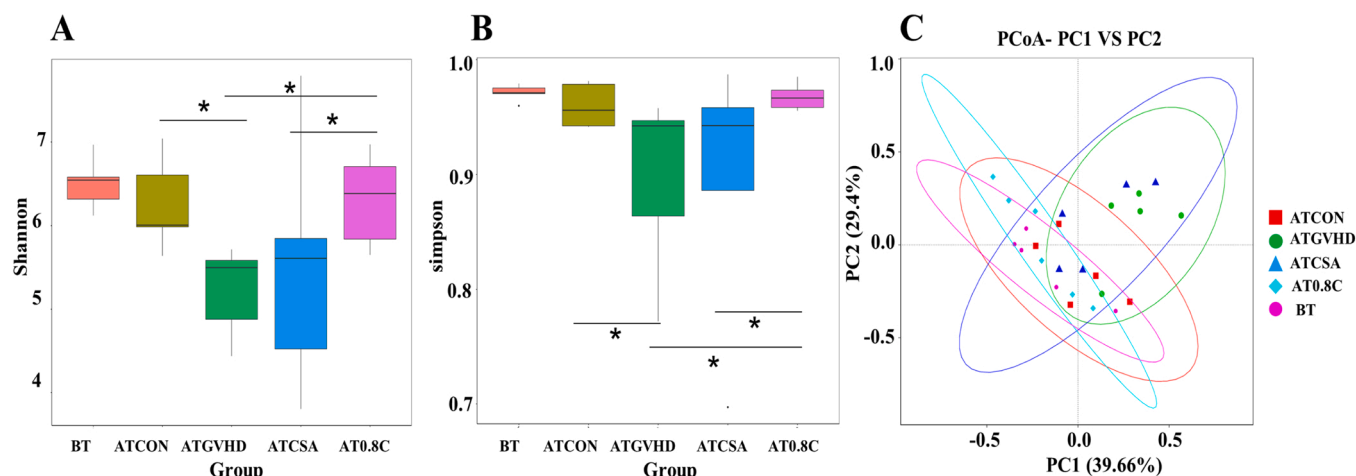


Fig. 3. C0127 combined with CSA maintained the diversity of gut microbiota. The Alpha diversity index Shannon (A) and Simpson (B) were presented by box plots. The comparison of microbiota changes in stool samples was analyzed by LefSe (linear discriminant analysis effect size). The lines represent the medians, the upper and lower limits of the box plot represent the 25th and 75th percentiles, and the error bars depict the 10th and 90th percentiles. The statistical significance was calculated by the Wilcoxon rank-sum test. (C) Principal component analysis (PCoA) of the unweighted Unifrac (a distance metric used to compare biological communities) distances. The meaning of abbreviation in figures: BT, before transplant (WT mice); ATCON, Control; ATCSA, 5 mg/kg CSA; AT 0.8 C, 0.8 mL/kg C0127 + 5 mg/kg CSA. * $P < 0.05$, compared with aGVHD mice.

whether C0127 could impact biomarkers of GVHD on species level. Compared the species in aGVHD mice with Control and C0127 + CSA group severally, we found both *Enterococcus durans* and *E. coli* were among the top 10 species in aGVHD mice via LefSe analysis while the relative abundances of two species in C0127 + CSA group was lower than aGVHD mice, similar to Control group (ATCON) (Fig. 5A–D). In addition, we conducted analysis of ternary phase diagram, and the results also confirmed that C0127 combined with CSA significantly reduced the abundances of *E. coli* and *Enterococcus durans* (Fig. 5E, F). Consistent with our results, other groups found that *E. coli* and *Enterococcus* contribute to the development of GVHD [9,58]. Besides, transplanting *E. coli* compromised the survival of aGVHD mice (Fig. 5G, H).

3.6. Protective effect of C0127 combined with 2.5 mg/kg CSA on aGVHD mice

To determine whether C0127 has a "synergistic and attenuated" effect on CSA, we reduced the dose of CSA and tested effect of 0.8 mL/kg C0127 and 2.5 mg/kg CSA on aGVHD mice. This treatment significantly improved the survival, body weight and GVHD score after transplant compared with model group (Suppl. Fig. 3A–E). In addition, HE morphology of liver tissues in C0127 combined with 2.5 mg/kg CSA group showed less damage, and its morphology was similar to control group (Suppl. Fig. 3F–I). Afterwards, it also found that comparing with aGVHD and CSA group, the intestinal injury was significantly reduced by combined administration (Suppl. Fig. 4). In conclusion, we revealed that C0127 could achieve a "synergistic and attenuated" effect of CSA in aGVHD mice.

3.7. XBJ/C0127 combined with CSA normalized gene expression in colon tissue of GVHD mice

To identify potential regulatory genes of XBJ/C0127 in the colon tissues of GVHD mice, RNA sequencing was performed to determine gene expression profiles in the colon tissues. Fig. 6A and Suppl. Fig. 2A showed the overall gene expression profiles of each mice in different groups. Fig. 6B revealed the differentially expressed genes (DEGs) counts between different groups. There were 1409 DEGs in the XBJ treatment group and 1729 DEGs in the C0127 treatment group with $\text{Log}_2\text{FC} > 1$ and $p\text{-value} < 0.05$ when comparing with aGVHD mice (Fig. 6C and D).

3.8. The combination of C0127 and CSA regulated key GVHD pathways and targets in the intestinal epithelium

The thresholds of DEGs were set to $\text{Log}_2\text{FC} > 1$ and $p\text{-value} < 0.05$, respectively, and the selected DEGs were analyzed by KEGG to determine the main regulatory signaling pathway of XBJ/C0127 in GVHD mice. The classic pathways were ranked based on the $-\log(p\text{-value})$ score, and the top 20 signaling pathways were listed (Fig. 7A and B). Notably, IL-17 signaling pathway ranked first in both rankings. Therefore, it was considered to be a key signaling pathway for XBJ/C0127 + CSA to regulate. There were 22 related genes in this pathway including G-CSF, CXCL2, CXCL3, CXCL5, S100A8, S100A9, LCN2, IL13 and so on (Fig. 7C and D). At the same time, the protein-protein interaction (PPI) analysis of related genes in the IL-17 signaling pathway revealed the important genes regulated by XBJ/C0127 + CSA (Fig. 7E and F). Through literature review and fold changes, 9 genes were identified as the major targets in IL-17 signaling and further verified.

3.9. XBJ/C0127 combined with CSA down-regulated the expression of cytokines/chemokines in the IL-17 signaling pathway

Next, we further validated the genes in the IL-17 signaling pathway. Fig. 8A showed the detailed diagram of IL-17 signaling pathway. RT-PCR was used to verify the accuracy of transcriptome results, and a total of 9 genes were verified, namely G-CSF, CXCL2, CXCL3, CXCL5, S100A8, S100A9, LCN2, IL13 and IL17RC (Fig. 8B–G) (the results of CXCL2, IL17RC and IL13 were shown in the Suppl. Fig. 5A–C). The results of RT-PCR were consistent with the transcriptome, that is, XBJ/C0127 + CSA could significantly down-regulate the expression of these cytokines/chemokines in IL-17 signaling pathway, which proved that the transcriptome data had high reliability and repeatability. G-CSF promotes neutrophil release from the BM into the circulation and its excessive release induces the gut inflammation and injury [1,46]. Therefore, our subsequent experiments mainly focused on G-CSF and its downstream signaling.

3.10. XBJ/C0127 inhibited G-CSF and STAT3 protein expression in gut tissues of aGVHD mice

According to relevant literature, STAT3 is a downstream target of G-CSF [61]. G-CSF can stimulate transcription factors, signal sensors,

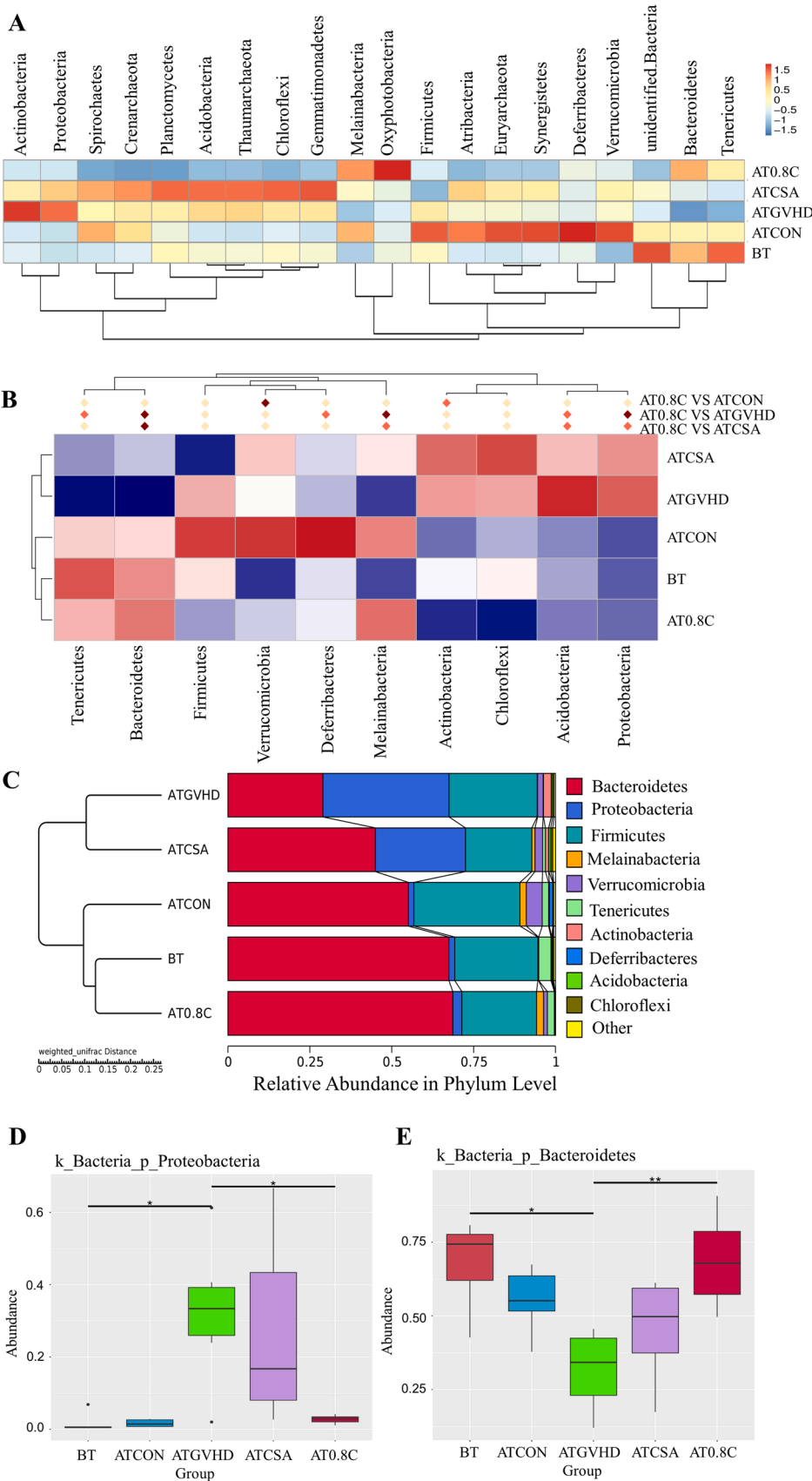
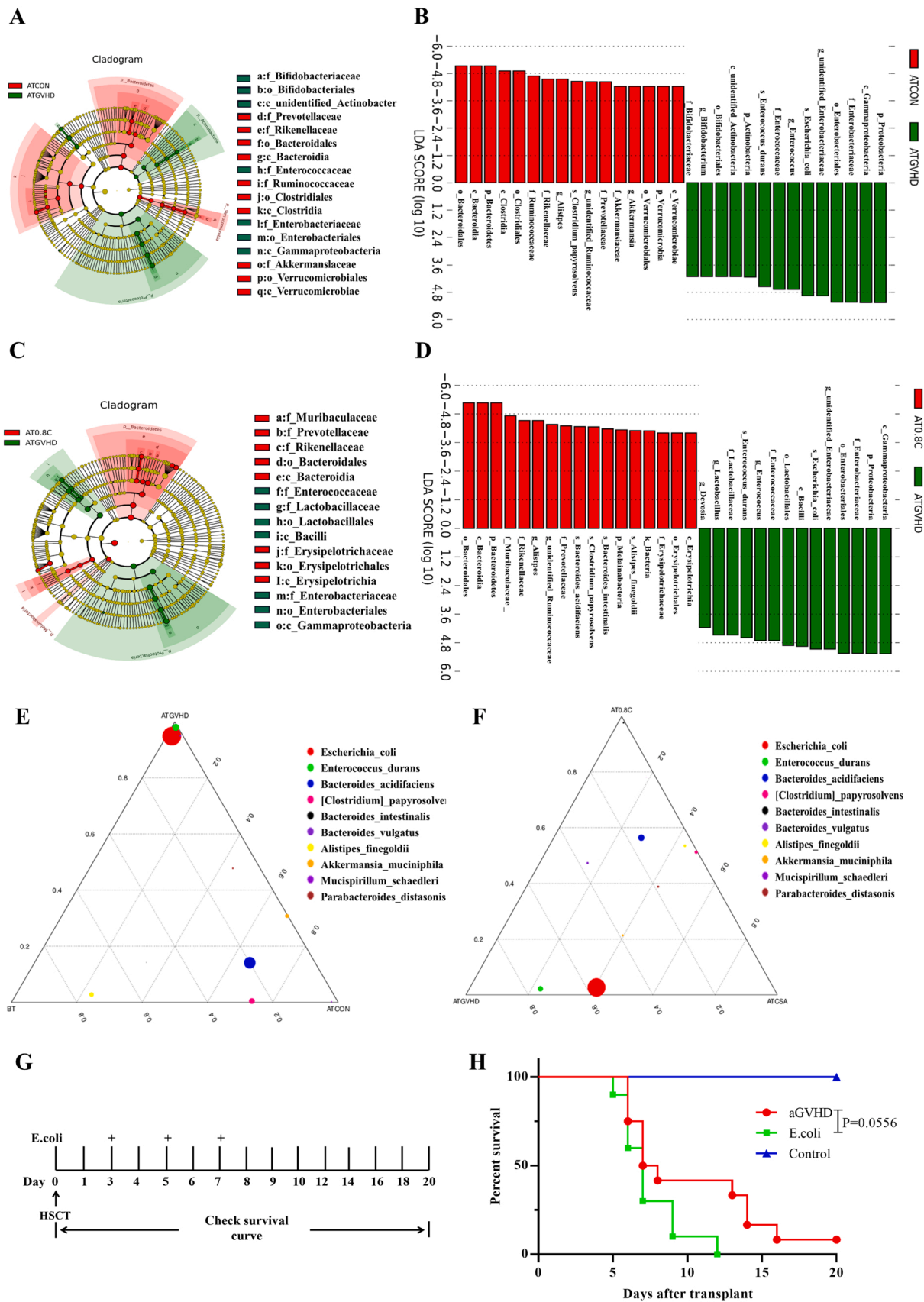


Fig. 4. (A) C0127 combined with CSA reversed the dysbiosis of aGVHD mice at phylum level. The heat map of the top 20 phylum of gut microbiota in different groups. The heat map is color-based on row Z-scores and the highest and lowest bacterial level are in red and blue, respectively. (B) The annotation heatmap of top 10 phylums were analyzed by MetaStat. The blue means down-regulation, red means up-regulation. (C) Unweighted Pair-group Method with Arithmetic Mean (UPGMA) tree based on Weighted UniFrac in phylum level presented the relative abundance of top 10 phylum. (D) The abundance of Bacteroidetes using MetaStat analysis and were shown by box plots in different groups. (E) The abundance of Proteobacteria using MetaStat analysis and were shown by box plots in different groups. n = 5–6/group, *P < 0.05, **P < 0.01, vs aGVHD group.



(caption on next page)

Fig. 5. C0127 combined with CSA reduced the abundance of *Escherichia coli* and *Enterococcus* in aGVHD mice. (A) The cladogram of Control (ATCON) and aGVHD (ATGVHD) mice. (B) LDA (Linear Discriminant Analysis) score generated by LEfSe (Linear Discriminant Analysis Effect Size) indicating differences in bacterial taxa. The central point represents the root of the tree (bacteria), and each ring represents the next lower taxonomic level (from phylum to genus). The diameter of each circle represents the relative abundance of the taxon. Only the taxa that significant LDA threshold value > 2 were shown. (C&D) The cladogram and LDA score of aGVHD and 5 mg/kg CSA + 0.8 mL/kg C0127 (AT 0.8 C). The species in Control (ATCON) and 5 mg/kg CSA + 0.8 mL/kg C0127 treatment group (AT 0.8 C) were indicated with a negative LDA score. (E&F) Analysis of ternary phase diagram in five groups including BT, ATCON, ATGVHD, ATCSA and AT 0.8 C. The distance between target point and angle represents the relative proportion of components in the corner group. The closer of the distance between the target point and angle, its proportion is higher in the corresponding group. (G) The experiment scheme of *E. coli* transplant. (H) The survival curve of control groups and aGVHD mice transplanted with *E. coli*.

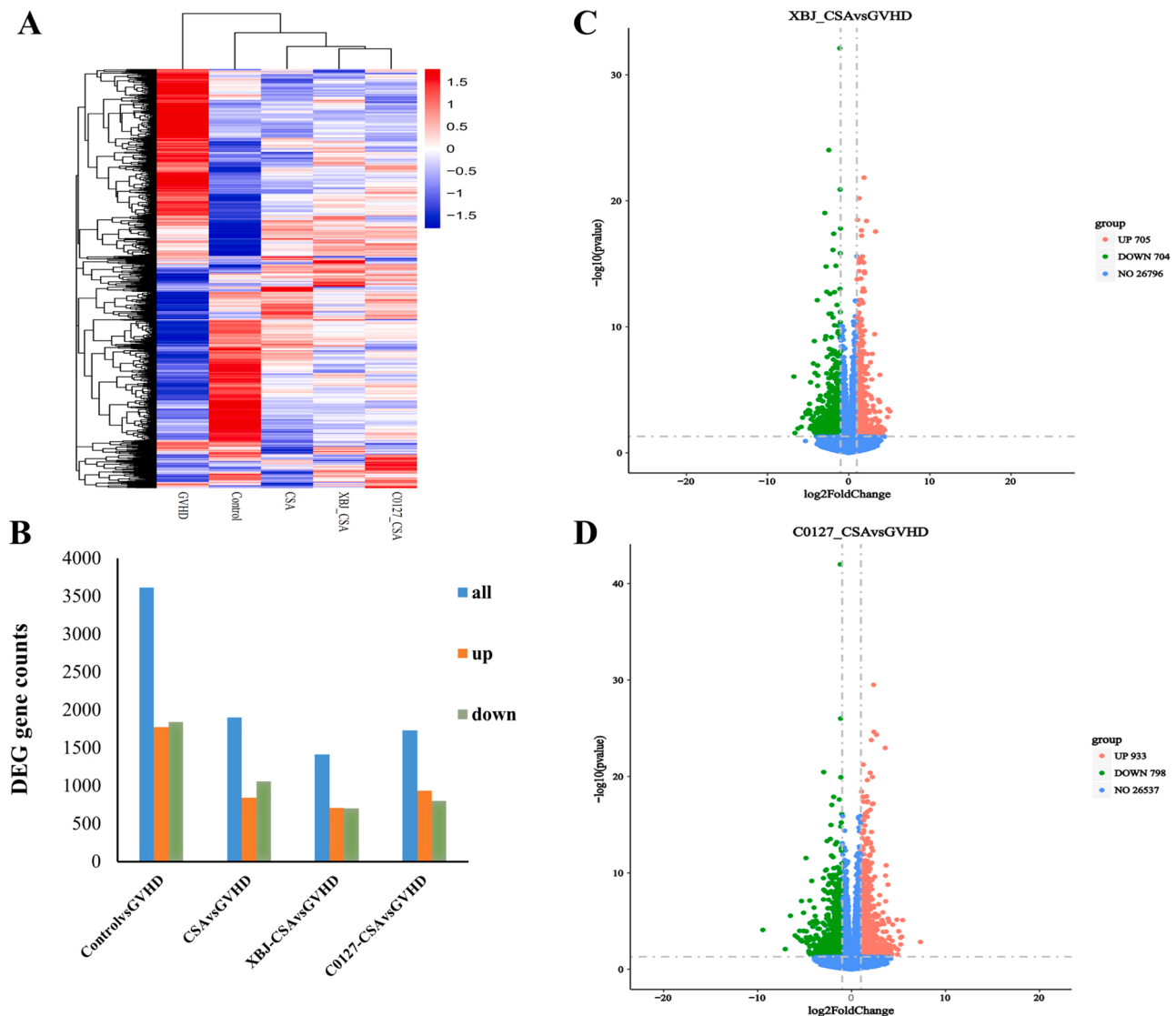


Fig. 6. XBJ/C0127 combined with CSA normalized the gene expression profile in the colon tissues of aGVHD mice. On the seventh day after transplantation, mice were euthanized and colon tissues were harvested and subjected to RNA-seq analysis. (A) Heat map of the gene expression profiles in different groups. (B) Differential gene counts between different groups. (C~D) Overall distribution of differentially expressed genes (DEGs) ($\log_2$ fold change > 1 and p-value < 0.05) was reflected by the volcano map. The X-axis represented the changes of gene expression in different samples and Y-axis represented the statistical significance of the difference in the gene expression. The up-regulated genes were red and down-regulated genes were green. (C) Differently expressed genes of XBJ+CSA vs GVHD group. (D) Differently expressed genes of C0127 + CSA vs GVHD group, n = 4/group.

catalysts of STAT1 and STAT5, and strongly activate STAT3 (Fig. 9A). The western blot assay showed that G-CSF expression significantly increased in the colon tissues of GVHD mice, while the combination treatment significantly reduced its level (Fig. 9B and C). In addition, the IHC assay showed that XBJ/C0127 + CSA reversed the abnormal elevation of G-CSF protein in colon tissues induced by GVHD (Fig. 9D and E). G-CSF production is induced by several inflammatory stimuli such as IL-1 β , TNF- α and lipopolysaccharide (LPS) [48]. Furthermore, *in*

vitro, we stimulated CACO2 cells with LPS to verify the effects of XBJ/C0127 on the expression of G-CSF and STAT3. RT-PCR results showed that XBJ and C0127 inhibited their expression on mRNA level (Fig. 9F and G).

4. Discussion

aGVHD remains a major life-threatening complication of allo-HSCT.

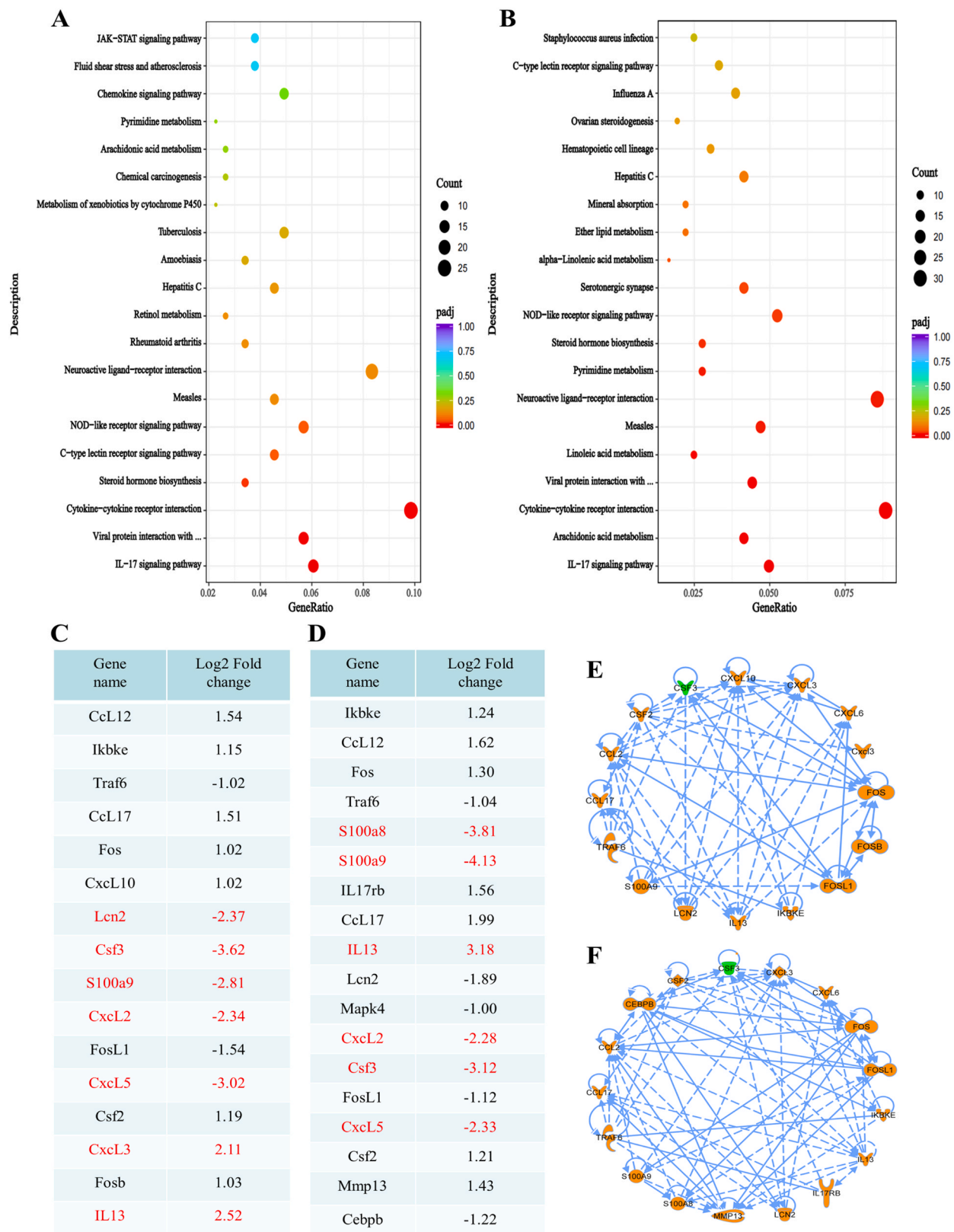


Fig. 7. XBJ/C0127 combined with CSA regulated critical pathways and targets in aGVHD mice. (A&B) KEGG analysis revealed the top 20 pathways of the DEGs. According to the $-\log(p\text{-value})$ score, IL-17 signaling was the top 1 pathway in XBJ (A) and C0127 (B) combined with CSA groups when comparing with GVHD group. (C&D) The lists of key targets and corresponding fold change in IL-17 signaling of XBJ+CSA (C) and C0127 + CSA (D), the verified genes were marked in red. (E&F) The protein-protein interaction (PPI) analysis of related genes in IL-17 signaling pathway of XBJ+CSA (E) and C0127 + CSA (F) groups, which revealed the important genes regulated by XBJ/C0127 + CSA.

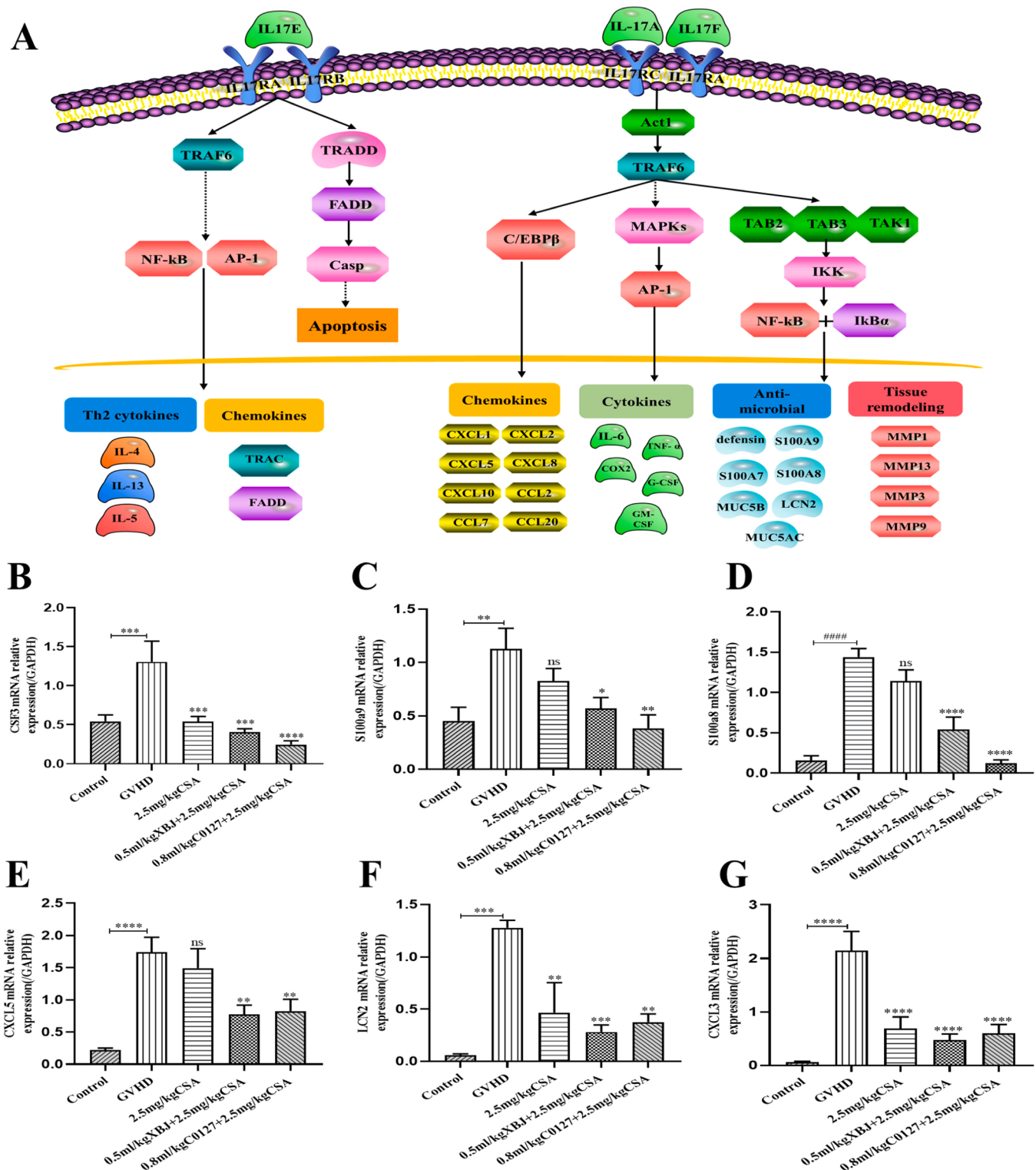


Fig. 8. XBJ/C0127 combined with CSA regulated the mRNA expression of IL-17 signaling pathway in the colon tissue of GVHD mice. (A) IL-17 signaling pathway. (B–G) The expression of chemokines in the colon tissues of different group mice, which were determined by RT-PCR, CSF3 (B), S100a9 (C), S100a8 (D), CXCL5 (E), LCN2 (F) and CXCL3 (G). Results were presented as mean $\pm$ SEM (n = 4/group). ns: no significance, *p < 0.05, **p < 0.01, ***p < 0.001, ****p < 0.0001, compared with GVHD group.

Herbal formulas have been used for thousands of years to treat human diseases globally. However, most formulations were not rigorously evaluated in adequate animal models and randomized clinical trials. TCM has multi-target and multi-component characteristics and has the potential to prevent and treat bacterial flora disorder. Our previous work

revealed that combining XBJ with CSA improved aGVHD prevention and ameliorated the gut damage of aGVHD mice [40] (Zhou et al., [65]). Thus, we further wondered which compounds play the main role and which signaling pathway regulated by XBJ in intestine. Here, we tested a formula with defined ingredients in a GVHD model, which shed light on

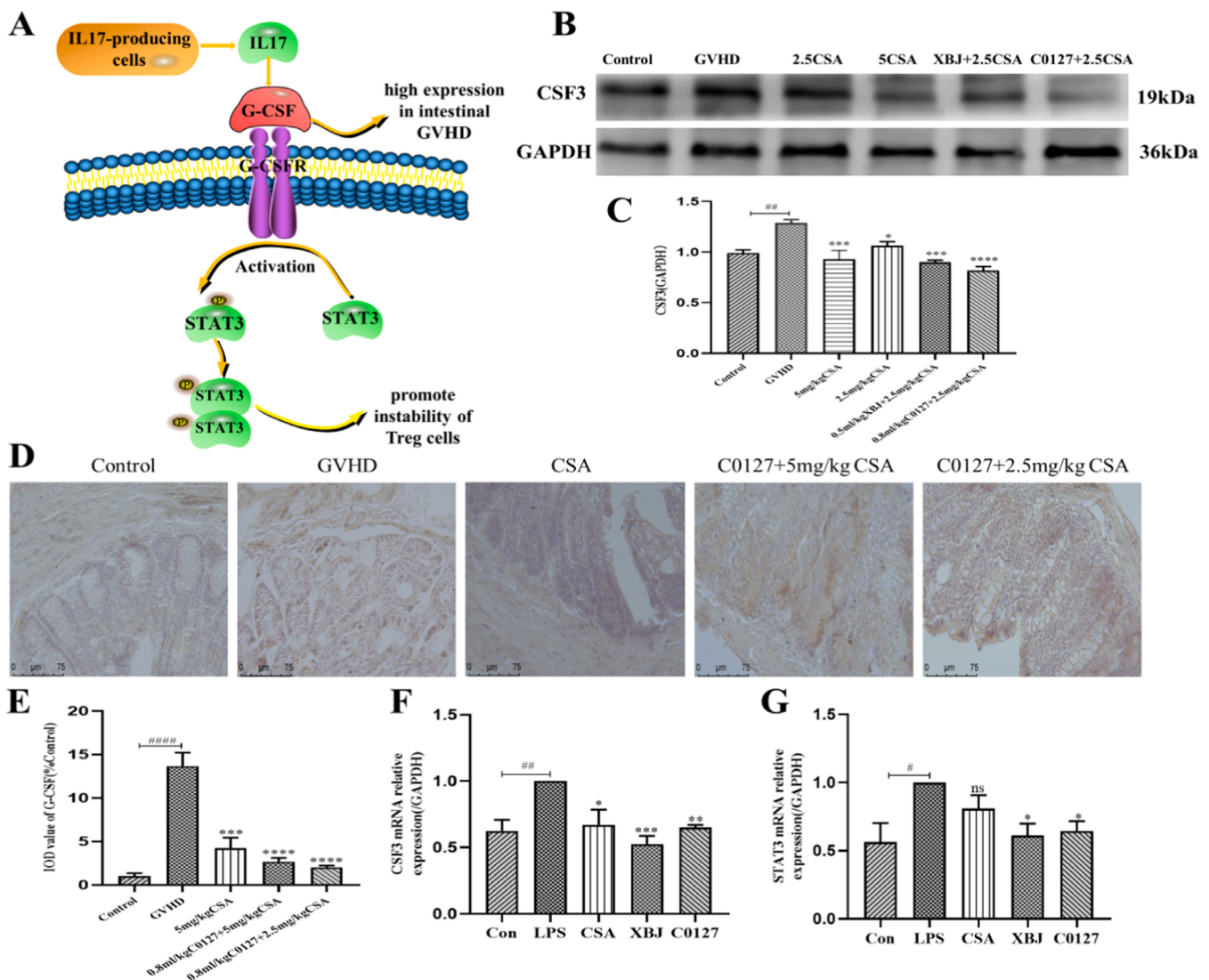


Fig. 9. XBJ/C0127 reduced the expression of G-CSF and STAT3 protein. (A) Upstream and downstream diagram of IL-17 and STAT3. (B) Representative western blot images of CSF3 protein expression in colon tissues after different treatments. (C) Quantification of G-CSF protein by Western blot ($n = 3$ per group). (D&E) The representative images and quantitative result of G-CSF expression in intestine tissues of each group mice using immunohistochemical technology. (F&G) The expression of G-CSF (F) and STAT3 (G) in CACO2 cells, which were detected by RT-PCR. Data are presented as mean $\pm$ SD values. ns: no significance, * $p < 0.05$, ** $p < 0.01$, compared with GVHD or LPS group.

improving the safety and efficacy of TCM. And transcriptomics analysis were used to determine the protective mechanisms of XBJ and C0127 on gut tissue of aGVHD mice.

In this work, we found combining four main compounds in XBJ (C0127) improved the preventive and treatment effects of CSA and produce the similar effects as XBJ mainly manifesting in ameliorating survival, weight loss, GVHD score, and inflammatory cytokines. Moreover, it has been reported that paeoniflorin, hydroxysafflor yellow A, protocatechualdehyde, and ferulic acid ameliorate organ injuries [31, 47,52,64].

Damage to the intestine results in the initiation and amplification of systemic inflammation and subsequent GVHD, consistently, the results of colon HE staining, FITC-dextran assay confirmed that C0127 combined with CSA reduced intestine damage caused by TBI and aGVHD (Fig. 2A–E). In addition, we found C0127 + CSA could maintain the SCFA level in feces (Fig. 2G–I). Meanwhile, recent studies showed that butyrate facilitates regulatory T (Treg) cells in the colon in mouse models and the increased amount of butyrate ameliorated gut GVHD [41,43]. Thereby, we hypothesized that the combination treatment may prevent the intestinal flora disorder by alleviating the gut injuries.

Coincidentally, the 16 S rRNA sequencing analysis revealed that the combined treatment maintained the bacterial diversity and normalized the composition of species. Besides, the combination treatment reduced the relative abundance of Proteobacteria and increased the Bacteroidetes in phylum level of aGVHD mice, and some researches revealed the reduced abundance of Proteobacteria and increased Bacteroidetes is beneficial for alleviating GVHD [28,60]. On the species level, we found increased abundance of *E. coli* and *Enterococcus durans* in aGVHD mice, which is consistent with previous works. C0127 + CSA significantly reduced their abundance. Transplanting *E. coli* compromised the survival of aGVHD mice treated with C0127 + CSA, indicating the significance of reversing dysbiosis on limiting the progression of aGVHD (Fig. 6G and H).

IL-17 is the founding member of a novel family of inflammatory cytokines and consists of multiple cytokine members with multiple cognate receptors. Research revealed inhibiting IL-17 could improve the outcomes of patients with psoriasisiform GVHD [14]. IL17A and IL-17F can induce proinflammatory genes expression, such as G-CSF, CXCL1, and CXCL2 [13]. Neutrophils and macrophages infiltration play important roles in GVHD deterioration, and G-CSF-mediated

inflammation induces gut injury and intestinal flora disturbance [1,4,63]. Controlling the feed-forward cascade of GVHD in the GI tract is a therapeutic approach to prevent GVHD [32]. Our transcriptome and RT-PCR results of intestine tissues uncovered XBJ/C0127 combined with CSA could reduce the expression of chemokines and cytokines mediating IL-17 signaling pathway. Furthermore, WB and IHC assays further confirmed that XBJ/C0127 + CSA decreased the G-CSF level and its downstream effector, STAT3. *In vitro*, XBJ and C0127 also significantly reduced the mRNA expression of G-CSF and STAT3 in CACO2 cells (Fig. 9F and G).

In conclusion, Combining C0127 with CSA restored the homeostasis of the intestinal microenvironment partially by normalizing the expression of G-CSF-STAT3 axis expression, resulting in the normalization of the gut microbiota. C0127 represents the main effective ingredients in XBJ to prevent aGVHD in mice. Our works provided a rationale to test the safety and effectiveness of C0127 in the clinic. However, the mechanism of C0127 on curbing the expansion of *E. coli* and *Enterococcus*, and how C0127 improves the effect of CSA remain to be studied.

Funding

This project was supported by the National Natural Science Foundation of China (Nos: 81774018, 81973581, 81873037); Tianjin Municipal Education Commission (Grant number: TD13-5046); Tianjin Municipal Science and Technology Bureau (No. 20ZY JDJC00070).

CRediT authorship contribution statement

Yuxin Feng designed and supervised the implementation of this research. Ting Shang and Yue Guo conducted experiments presented in all Figures. Ting Shang, Yue Guo, Xiu-Rong Li, Zhengcan Zhou, and Yubo Qi conducted experiments presented in Fig. 1 and Supplement Figs. 1,2. Ting Shang conducted experiments presented in Figs. 2–5. Yue Guo conducted experiments presented in Figs. 6–9. Yue Guo, Ru Shen conducted experiments presented in Supplement Fig. 3. Shuang He, Mei Wang, Zhe-Xin Shi, Guanwei Fan, Yuefei Wang, Xiumei Gao and Yan Zhu participated in the analysis and interpretation of the results. Khalid Salahdiin, Xin Zhao and Jian Yang participated in revising the manuscript. Yuxin Feng, Ting Shang, and Yue Guo wrote the manuscript. All authors reviewed and approved the manuscript.

Conflict of interest statement

The authors declare no conflict of interest.

Acknowledgments

We would like to thank our colleagues, particularly Pengzhi Dong, Rui Shao, and John Owoicho Orgah, as well as Xintong Wang, Yule-Wang, Zhixiong Li and Guangxu Xiao for stimulating discussion and sharing reagents.

Author contributions

TS, YG, YXF, and YZ designed the study and developed the methodologies. TS, YG, XRL, ZCZ, YBQ, RS and YXF conducted research. TS, YG, KS, SH, XZ, ZXS, MW, JY, GWF, YFW, XMG and YXF analyzed the data, reviewed the manuscript, and contributed critical reagents. TS, YG and YXF wrote and revised the manuscript.

Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.biopha.2022.112675](https://doi.org/10.1016/j.biopha.2022.112675).

References

- [1] L. Ai, Y. Ren, M. Zhu, S. Lu, Y. Qian, Z. Chen, A. Xu, Synbindin restrains proinflammatory macrophage activation against microbiota and mucosal inflammation during colitis, *Gut* 70 (2021) 2261–2272, [utjnl-2020-321094](https://doi.org/10.1136/gut-2020-321094).
- [2] A.S. Al-Homsi, A. Goodyke, M. McLane, S. Abdel-Mageed, K. Cole, M. Muilenburg, Y. Feng, Post-Transplantation cyclophosphamide and ixazomib combination rescues mice subjected to experimental graft-versus-host disease and is superior to either agent alone, *Biol. Blood Marrow Transpl.* 23 (2017) 255–261.
- [3] T.M. Andermann, J.U. Peled, C. Ho, P. Reddy, M. Riches, R. Storb, T. Teshima, M.R. M. van den Brink, A. Alousi, S. Balderman, P. Chiusolo, W.B. Clark, E. Holler, A. Howard, L.S. Kean, A.Y. Koh, P.L. McCarthy, J.M. McCarty, M. Mohty, R. Nakamura, K. Rezvani, B.H. Segal, B.E. Shaw, E.J. Shpall, A.D. Sung, D. Weber, J. Whangbo, J.R. Wingard, W.A. Wood, M.A. Perales, R.R. Jenq, A.S. Bhatt, N. Blood and Marrow Transplant Clinical Trials, The microbiome and hematopoietic cell transplantation: past, present, and future, *Biol. Blood Marrow Transpl.* 24 (2018) 1322–1340.
- [4] S. An, I. Raju, B. Surenkhuu, J.E. Kwon, S. Gulati, M. Karaman, A. Pradeep, S. Sinha, C. Mun, S. Jain, Neutrophil extracellular traps (NETs) contribute to pathological changes of ocular graft-vs.-host disease (oGVHD) dry eye: implications for novel biomarkers and therapeutic strategies, *Ocul. Surf.* 17 (3) (2019) 589–614.
- [5] M.B. Baker, N.H. Altman, E.R. Podack, R.B. Levy, The role of cell-mediated cytotoxicity in acute GVHD after MHC-matched allogeneic bone marrow transplantation in mice, *J. Exp. Med.* 183 (1996) 2645–2656.
- [6] A. Beringer, P. Miossec, Systemic effects of IL-17 in inflammatory, *Arthritis Nat. Rev. Rheumatol.* 15 (8) (2019) 491–501.
- [7] A.S. Bhatt, S.S. Freeman, A.F. Herrera, C.S. Pedamallu, D. Gevers, F. Duke, J. Jung, M. Michaud, B.J. Walker, S. Young, A.M. Earl, A.D. Kostic, A.I. Ojesina, R. Hasslerjian, K.K. Ballen, Y.B. Chen, G. Hobbs, J.H. Antin, R.J. Soiffer, L.R. Baden, W.S. Garrett, J.L. Hornick, F.M. Marty, M. Meyerson, Sequence-based discovery of *Bradyrhizobium enterica* in cord colitis syndrome, *N. Engl. J. Med.* 369 (2013) 517–528.
- [8] K.L. Bowerman, A. Varelias, N. Lachner, R.D. Kuns, G.R. Hill, P. Hugenholtz, Continuous pre- and post-transplant exposure to a disease-associated gut microbiome promotes hyper-acute graft-versus-host disease in wild-type mice, *Gut Microbes* 11 (2020) 754–770.
- [9] Y.J. Chang, X.Y. Zhao, X.J. Huang, Granulocyte colony-stimulating factor-primed unmanipulated haploidentical blood and marrow transplantation, *Front. Immunol.* 10 (2019) 2516.
- [10] Y. Chen, H. Tong, Z. Pan, D. Jiang, X. Zhang, J. Qiu, L. Su, M. Zhang, Xuebijing injection attenuates pulmonary injury by reducing oxidative stress and proinflammatory damage in rats with heat stroke, *Exp. Ther. Med.* 13 (2017) 3408–3416.
- [11] C. Cheng, J.Z. Lin, L. Li, J.L. Yang, W.W. Jia, Y.H. Huang, F.F. Du, F.Q. Wang, M. J. Li, Y.F. Li, F. Xu, N.T. Zhang, O.E. Olaleye, Y. Sun, J. Li, C.H. Sun, G.P. Zhang, C. Li, Pharmacokinetics and disposition of monoterpane glycosides derived from *Paeonia lactiflora* roots (Chishao) after intravenous dosing of antiseptic XueBiJing injection in human subjects and rats, *Acta Pharm. Sin.* 37 (2016) 530–544.
- [12] S.B. Coffelt, K. Kersten, C.W. Doornebal, J. Weiden, M. Vrijland, C.S. Hau, N.J. M. Verstegen, M. Ciampricotti, L. Hawinkels, J. Jonkers, K.E. de Visser, IL-17-producing $\gamma\delta$ T cells and neutrophils conspire to promote breast cancer metastasis, *Nature* 522 (7556) (2015) 345–348.
- [13] P.E. Debureaux, A. de Masson, M. Battistella, F. Sicre de Fontbrune, G. Socié, J. D. Bouaziz, D. Michonneau, Chronic graft-versus-host disease and inhibition of interleukin-17: proof of concept in humans, *Br. J. Dermatol.* 182 (4) (2020) 1038–1041.
- [14] Y. Dong, D. Zhang, Q. Yu, Q. Zhao, C. Xiao, K. Zhang, C. Jia, S. Chen, B. Zhang, B. Zhang, M. Li, Loss of Ssq1 leads to mitochondrial dysfunction, activation of autophagy and cell cycle arrest due to iron overload triggered by mitochondrial iron-sulfur cluster assembly defects in *Candida albicans*, *Int. J. Biochem. Cell Biol.* 85 (2017) 44–55.
- [15] R.C. Edgar, Muscle: multiple sequence alignment with high accuracy and high throughput, *Nucleic Acids Res.* 32 (2004) 1792–1797.
- [16] R.C. Edgar, UPARSE: highly accurate OTU sequences from microbial amplicon reads, *Nat. Methods* 10 (2013) 996–998.
- [17] R.C. Edgar, B.J. Haas, J.C. Clemente, C. Quince, R. Knight, UCHIME improves sensitivity and speed of chimera detection, *Bioinformatics* 27 (2011) 2194–2200.
- [18] J.L. Ferrara, J.E. Levine, P. Reddy, E. Holler, Graft-versus-host disease, *Lancet* 373 (2009) 1550–1561.
- [19] J. Gao, L. Kong, S. Liu, Z. Feng, H. Shen, Q. Liu, A prospective multicenter clinical study of Xuebijing injection in the treatment of sepsis and multiple organ dysfunction syndrome, *Zhonghua Wei Zhong Bing Ji Jiu Yi Xue*, 27, 2015, pp. 465–470.
- [20] K.H. Gartlan, H. Bommasamy, K. Paz, A.N. Wilkinson, M. Owen, D. K. Reichenbach, T. Banovic, K. Wehner, F. Buchanan, A. Varelias, R.D. Kuns, K. Chang, Y. Fedoriv, T. Shea, J. Coghill, M. Zaiken, M.W. Plank, P.S. Foster, A. D. Clouston, B.R. Blazar, J.S. Serody, G.R. Hill, A critical role for donor-derived IL-22 in cutaneous chronic GVHD, *Am. J. Transpl.* 18 (4) (2018) 810–820.
- [21] A.P. Gobert, N.T. Al-Greene, K. Singh, L.A. Coburn, J.C. Sierra, T.G. Verriere, P. B. Luis, C. Schneider, M. Asim, M.M. Allaman, D.P. Barry, J.L. Cleveland, C. E. Destefano Shields, Jr Casero RA, M.K. Washington, M.B. Piazuelo, K.T. Wilson, Distinct immunomodulatory effects of spermine oxidase in colitis induced by epithelial injury or infection, *Front. Immunol.* 9 (2018) 1242.
- [22] M. Gooptu, J.H. Antin, GVHD prophylaxis 2020, *Front. Immunol.* 12 (2021), 605726.

- [24] B.J. Haas, D. Gevers, A.M. Earl, M. Feldgarden, D.V. Ward, G. Giannoukos, D. Ciulla, D. Tabbaa, S.K. Highlander, E. Sodergren, B. Methé, T.Z. DeSantis, C. Human Microbiome, J.F. Petrosino, R. Knight, B.W. Birren, Chimeric 16S rRNA sequence formation and detection in Sanger and 454-pyrosequenced PCR amplicons, *Genome Res.* 21 (2011) 494–504.
- [25] A.C. Harris, R. Young, S. Devine, W.J. Hogan, F. Ayuk, U. Bunworasate, C. Chanswangphuwana, Y.A. Efebera, E. Holler, M. Litzow, R. Ordemann, M. Qayed, A.S. Renteria, R. Reshef, M. Wölfl, Y.B. Chen, S. Goldstein, M. Jagasia, F. Locatelli, S. Mielke, D. Porter, T. Schechter, Z. Shekhovtsova, J.L. Ferrara, J. E. Levine, International, multicenter standardization of acute graft-versus-host disease clinical data collection: a report from the Mount Sinai Acute GVHD International Consortium, *Biol. Blood Marrow Transpl.* 22 (2016) 4–10.
- [26] E. Holler, P. Butzhammer, K. Schmid, C. Hundsruker, J. Koestler, K. Peter, W. Zhu, D. Sporer, T. Hehlhans, M. Kreutz, B. Holler, D. Wolff, M. Edinger, R. Andreesen, J. E. Levine, J.L. Ferrara, A. Gessner, R. Spang, P.J. Oefner, Metagenomic analysis of the stool microbiome in patients receiving allogeneic stem cell transplantation: loss of diversity is associated with use of systemic antibiotics and more pronounced in gastrointestinal graft-versus-host disease, *Biol. Blood Marrow Transpl.* 20 (2014) 640–645.
- [27] H. Huang, L. Ji, S. Song, J. Wang, N. Wei, M. Jiang, G. Bai, G. Luo, Identification of the major constituents in Xuebijing injection by HPLC-ESI-MS, *Phytochem. Anal.* 22 (2011) 330–338.
- [28] E.L. Johnson, S.L. Heaver, W.A. Walters, R.E. Ley, Microbiome and metabolic disease: revisiting the bacterial phylum bacteroidetes, *J. Mol. Med.* 95 (2017) 1–8.
- [29] L.T. Khuat, C.T. Le, C.S. Pai, R.R. Shields-Cutler, S.G. Holtan, A. Rashidi, S. L. Parker, D. Knights, J.I. Luna, C. Dunai, Z. Wang, I.R. Sturgill, K.M. Stoffel, A. A. Merleev, S.K. More, E. Maverakis, H.E. Raybould, M. Chen, R.J. Canter, A. M. Monjaze, M. Dave, J. Ferrara, J.E. Levine, D.L. Longo, M. Abedi, B.R. Blazar, W.J. Murphy, Obesity induces gut microbiota alterations and augments acute graft-versus-host disease after allogeneic stem cell transplantation, *Sci. Transl. Med.* 12 (571) (2020) eaay7713.
- [30] E. Kolaczowska, P. Kubes, Neutrophil recruitment and function in health and inflammation, *Nat. Rev. Immunol.* 13 (3) (2013) 159–175.
- [31] B.S. Kong, Y.H. Cho, E.J. Lee, G protein-coupled estrogen receptor-1 is involved in the protective effect of protocatechuic aldehyde against endothelial dysfunction, *PLoS One* 9 (11) (2014), e113242.
- [32] M. Koyama, G.R. Hill, Alloantigen presentation and graft-versus-host disease: fuel for the fire, *Blood* 127 (24) (2016) 2963–2970.
- [33] J. Lee, J. d'Aigle, L. Atadja, V. Quaicoe, P. Honarpisheh, B.P. Ganesh, A. Hassan, J. Graf, J. Petrosino, N. Putluri, L. Zhu, D.J. Durgan, Jr Bryan RM, L. D. McCullough, V.R. Venna, Gut microbiota-derived short-chain fatty acids promote poststroke recovery in aged mice, *Circ. Res.* 127 (4) (2020) 453–465.
- [34] C. Li, G. Ai, Y. Wang, Q. Lu, C. Luo, L. Tan, G. Lin, Y. Liu, Y. Li, H. Zeng, J. Chen, Z. Lin, Y. Xian, X. Huang, J. Xie, Z. Su, Oxyberberine, a novel gut microbiota mediated metabolite of berberine, possesses superior anti-colitis effect: impact on intestinal epithelial barrier, gut microbiota profile and TLR4-MyD88-NF- κ B pathway, *Pharmacol. Res.* 152 (2019), 104603.
- [35] G. Li, H. Chen, L. Liu, P. Xiao, Y. Xie, X. Geng, T. Zhang, Y. Zhang, T. Lu, H. Tan, L. Li, B. Sun, Role of interleukin-17 in acute pancreatitis, *Front. Immunol.* 12 (2021), 674803.
- [36] X. Li, R. Bechara, J. Zhao, M.J. McGeachy, S.L. Gaffen, IL-17 receptor-based signaling and implications for disease, *Nat. Immunol.* 20 (12) (2019) 1594–1602.
- [37] X. Li, C. Cheng, F. Wang, Y. Huang, W. Jia, O.E. Olaleye, M. Li, Y. Li, C. Li, Pharmacokinetics of catechols in human subjects intravenously receiving XueBiJing injection, an emerging antiseptic herbal medicine, *Drug Metab. Pharmacokinet.* 31 (2016) 95–98.
- [38] Y.C. Liu, F.H. Yao, Y.F. Chai, N. Dong, Z.Y. Sheng, Y.M. Yao, Xuebijing injection promotes M2 polarization of macrophages and improves survival rate in septic mice, *Evid. Based Complement. Altern. Med.* 2015 (2015), 352642.
- [39] M. Lyu, Z.C. Zhou, X.M. Wang, H. Lv, M. Wang, G.X. Pan, Y. Wang, G.W. Fan, X. M. Gao, Y.X. Feng, Y. Zhu, Network pharmacology-guided development of a novel integrative regimen to prevent acute graft-versus-host disease, *Front. Pharmacol.* 9 (2018) 1440.
- [40] K.A. Markey, J. Schluter, A.L.C. Gomes, E.R. Littmann, A.J. Pickard, B.P. Taylor, P. A. Giardina, D. Weber, A. Dai, M.D. Docampo, G.K. Armijo, A.E. Slingerland, J. B. Slingerland, K.B. Nichols, D.G. Brereton, A.G. Clurman, R.J. Ramos, A. Rao, A. Bush, L. Bohannon, M. Covington, M.V. Lew, D.A. Rizzieri, N. Chao, M. Maloy, C. Cho, I. Politikos, S. Giralt, Y. Taur, E.G. Pamer, E. Holler, M.A. Perales, D. M. Ponce, S.M. Devlin, J. Xavier, A.D. Sung, J.U. Peled, J.R. Cross, M.R.M. van den Brink, The microbe-derived short-chain fatty acids butyrate and propionate are associated with protection from chronic GVHD, *Blood* 136 (1) (2020) 130–136. Jul 2.
- [41] M. Martin, Cutadapt removes adapter sequences from high-throughput sequencing reads, *EMBnet J.* 17 (2011) 10–12.
- [42] N.D. Mathewson, R. Jenq, A.V. Mathew, M. Koenigsnecht, A. Hanash, T. Toubai, K. Oravec-Wilson, S.R. Wu, Y. Sun, C. Rossi, H. Fujiwara, J. Byun, Y. Shono, C. Lindemans, M. Calafiore, T.M. Schmidt, K. Honda, V.B. Young, S. Pennathur, M. van den Brink, P. Reddy, Gut microbiome-derived metabolites modulate intestinal epithelial cell damage and mitigate graft-versus-host disease, *Nat. Immunol.* 17 (2016) 505–513.
- [43] T. Matsumoto, M. Moriya, M.H. Sakurai, H. Kiyohara, Y. Tabuchi, H. Yamada, Stimulatory effect of a pectic polysaccharide from a medicinal herb, the roots of *Bupleurum falcatum* L., on G-CSF secretion from intestinal epithelial cells, *Int. Immunopharmacol.* 8 (4) (2008) 581–588.
- [44] J. Mei, Y. Liu, N. Dai, C. Hoffmann, K.M. Hudock, P. Zhang, S.H. Guttentag, J. K. Kolls, P.M. Oliver, F.D. Bushman, G.S. Worthen, Cxcr2 and Cxcl5 regulate the IL-17/G-CSF axis and neutrophil homeostasis in mice, *J. Clin. Investig.* 122 (3) (2012) 974–986.
- [45] F. Min, H. Sun, B. Wang, N. Ahmad, H. Guo, H. Gao, Y. Gao, X. Liu, H. Li, Hepatoprotective effects of hydroxysafflor yellow A in D-galactose-treated aging mice, *Eur. J. Pharmacol.* 881 (2020), 173214.
- [46] A.D. Panopoulos, S.S. Watowich, Granulocyte colony-stimulating factor: molecular mechanisms of action during steady state and 'emergency' hematopoiesis, *Cytokine* 42 (3) (2008) 277–288.
- [47] M.C. Pasquini, X. Zhu, Current Use and Outcome of Hematopoietic Stem Cell Transplantation, CIBMTR Summary Slides, 2015. (<https://www.cibmtr.org/Pages/index.aspx>).
- [48] C. Quast, E. Priesse, P. Yilmaz, J. Gerken, T. Schweer, P. Yazar, J. Peplies, F. O. Glockner, The SILVA ribosomal RNA gene database project: improved data processing and web-based tools, *Nucleic Acids Res.* 41 (2013) 590–596.
- [49] M.G. Rooks, W.S. Garrett, Gut microbiota, metabolites and host immunity, *Nat. Rev. Immunol.* 16 (2016) 341–352.
- [50] S. Sanjeev, R.M. Bidanchi, M.K. Murthy, G. Gurusubramanian, V.K. Roy, Influence of ferulic acid consumption in ameliorating the cadmium-induced liver and renal oxidative damage in rats, *Environ. Sci. Pollut. Res. Int.* 26 (20) (2019) 20631–20653.
- [51] T. Shang, Q. Yu, T. Ren, X.T. Wang, H. Zhu, J.M. Gao, G. Pan, X. Gao, Y. Zhu, Y. Feng, M.C. Li, Xuebijing injection maintains GRP78 expression to prevent *Candida albicans*-induced epithelial death in the kidney, *Front. Pharmacol.* 10 (2020) 1416.
- [52] H. Shi, Y. Hong, J. Qian, X. Cai, S. Chen, Xuebijing in the treatment of patients with sepsis, *Am. J. Emerg. Med.* 35 (2017) 285–291.
- [53] Y. Shono, M.R.M. van den Brink, Gut microbiota injury in allogeneic haematopoietic stem cell transplantation, *Nat. Rev. Cancer* 18 (5) (2018) 283–295.
- [54] M.H. Sofi, Y. Wu, T. Ticer, S. Schutt, D. Bastian, H.J. Choi, L. Tian, C. Mealer, C. Liu, C. Westwater, K.E. Armeson, A.V. Alekseyenko, X.Z. Yu, A single strain of *Bacteroides fragilis* protects gut integrity and reduces GVHD, *JCI Insight* 6 (3) (2021), e136841.
- [55] A. Staffas, M. Burgos da Silva, M.R. van den Brink, The intestinal microbiota in allogeneic hematopoietic cell transplant and graft-versus-host disease, *Blood* 129 (8) (2017) 927–933.
- [56] C.K. Stein-Thoeriger, K.B. Nichols, A. Lazrak, M.D. Docampo, A.E. Slingerland, J. B. Slingerland, A.G. Clurman, G. Armijo, A. Gomes, Y. Shono, A. Staffas, M. Burgos da Silva, S.M. Devlin, K.A. Markey, D. Bajic, R. Pinedo, A. Tsakmakis, E. R. Littmann, A. Pastore, Y. Taur, S. Monette, M.E. Arcila, A.J. Pickard, M. Maloy, R. J. Wright, L.A. Amoretti, E. Fontana, D. Pham, M.A. Jamal, D. Weber, A.D. Sung, D. Hashimoto, C. Scheid, J.B. Xavier, J.A. Messina, K. Romero, M. Lew, A. Bush, L. Bohannon, K. Hayasaka, Y. Hasegawa, M. Vehreschild, J.R. Cross, D.M. Ponce, M.A. Perales, S.A. Giral, R.R. Jenq, T. Teshima, E. Holler, N.J. Chao, E.G. Pamer, J. U. Peled, M. van den Brink, Lactose drives *Enterococcus* expansion to promote graft-versus-host disease, *Science* 366 (2019) 1143–1149.
- [57] Z. Sun, L. Zuo, T. Sun, J. Tang, D. Ding, L. Zhou, J. Kang, X. Zhang, Chemical profiling and quantification of Xuebijing injection, a systematic quality control strategy using UHPLC-Q exactive hybrid quadrupole-orbitrap high-resolution mass spectrometry, *Sci. Rep.* 7 (2017) 16921.
- [58] Y. Taur, J.B. Xavier, L. Lipuma, C. Ubeda, J. Goldberg, A. Gbourne, Y.J. Lee, K. A. Dubin, N.D. Succi, A. Viale, M.A. Perales, R.R. Jenq, M.R. van den Brink, E. G. Pamer, Intestinal domination and the risk of bacteremia in patients undergoing allogeneic hematopoietic stem cell transplantation, *Clin. Infect. Dis.* 55 (2012) 905–914.
- [59] X. Wang, Q. Chu, X. Jiang, Y. Yu, L. Wang, Y. Cui, J. Lu, L. Teng, D. Wang, *Sarcodon imbricatus* polysaccharides improve mouse hematopoietic function after cyclophosphamide-induced damage via G-CSF mediated JAK2/STAT3 pathway, *Cell Death Dis.* 9 (6) (2018) 578.
- [60] X.T. Wang, Z. Peng, Y.Y. An, T. Shang, G. Xiao, S. He, X. Chen, H. Zhang, Y. Wang, T. Wang, J.H. Zhang, X.M. Gao, Y. Zhu, Y. Feng, Paeoniflorin and hydroxysafflor yellow A in xuebijing injection attenuate sepsis-induced cardiac dysfunction and inhibit proinflammatory cytokine production, *Front. Pharmacol.* 11 (2021), 614024.
- [61] K. Wu, Y. Yuan, H. Yu, X. Dai, S. Wang, Z. Sun, F. Wang, H. Fei, Q. Lin, H. Jiang, T. Chen, The gut microbial metabolite trimethylamine N-oxide aggravates GVHD by inducing M1 macrophage polarization in mice, *Blood* 136 (4) (2020) 501–515.
- [62] L. Zhang, B. Yang, B. Yu, Paeoniflorin protects against nonalcoholic fatty liver disease induced by a high-fat diet in mice, *Biol. Pharm. Bull.* 38 (7) (2015) 1005–1011.
- [63] Z. Zhou, T. Shang, X. Li, Protecting intestinal microenvironment alleviates acute graft-versus-host disease, *Front. Physiol.* 11 (2021), 608279.